Health, Nutrition & Lifestyle Management System

INTEGRATED HEALTH DATA ANALYSIS

PROJECT OVERVIEW

This project combines health, nutrition, sleep, and physical activity data from multiple sources into one unified system. It aims to analyze lifestyle patterns and discover relationships between nutrition, sleep quality, stress levels, and physical activity using integrated health datasets.

- Nutrition
- Sleep
- Physical Activity

TEAM MEMBERS

- Lamar Abdel Aziz Alqurashi — 446003107 · G5
- Jana Abdul Rauf Saati — 446017177 · G5
- Fatima Shaker Alharthi — 446000667 · G7

Lab 1: Scenario & Data Sources

Many people struggle to maintain a healthy lifestyle due to poor sleep habits, unhealthy nutrition, high stress levels, and lack of physical activity.

Since health-related data is usually stored in separate systems and formats, it becomes difficult to analyze overall lifestyle patterns effectively.

This project aims to integrate multiple health and lifestyle datasets into one unified system to support better lifestyle understanding and analysis.

DATA SOURCES

DATASET	TOTAL COLUMNS	DATA DESCRIPTION	INTEGRATION & RELATIONSHIP
Sleep Health & Lifestyle	13	Measures sleep quality, duration, and stress levels, along with demographic data such as age, gender, and occupation.	Connected with Fitbit activity data to analyze the relationship between sleep and physical activity.
Daily Food & Nutrition	12	Tracks daily meals and nutritional components consumed by the user, including calories, protein, fats, and carbohydrates.	Linked with sleep and lifestyle data to study the impact of nutrition on health patterns.
Fitbit Daily Activity	15	Records physical activity and health metrics from smart wearables, including daily steps, distance covered, and calories burned.	Integrated with sleep data to evaluate activity levels, calories burned, and sleep quality.
Food Reviews	5	Contains textual user feedback and ratings to serve as the unstructured layer for sentiment analysis.	Transformed into sentiment scores and linked with nutrition data to analyze food satisfaction and behavior.

Lab 2: Structured vs. Unstructured Data

DATA CLASSIFICATION

DATA SOURCE	TYPE	EXAMPLE
Sleep Health & Lifestyle	Structured	Sleep Duration, Stress Level
Fitbit Daily Activity	Structured	Daily Steps, Calories
Daily Food & Nutrition	Structured	Meal Type, Protein
Food Review	Unstructured	Review Text

STRUCTURED DATA

- Organized in rows and columns
- Easy to store and query using SQL

UNSTRUCTURED DATA

- Contains free text reviews
- Required preprocessing and sentiment analysis

Lab 3: Data Cleaning & Preprocessing

ISSUES & SOLUTIONS

ATTRIBUTE	PROBLEM TYPE	ISSUE DESCRIPTION	IMPACT BEFORE CLEANING	BEST SOLUTION
Blood Pressure	Invalid Data Type / Format	Stored as a text format (120/80)	Cannot perform separate numerical analysis	Split into systolic and diastolic columns
All attributes	Duplicate	Duplicate rows detected	Causes redundancy and biased analysis	Remove duplicates
Review Text	Unstructured	Free-text reviews with inconsistent styles	Difficult text analyze and poor NLP accuracy	Text preprocessing and sentiment analysis
Activity Date	Invalid Data Type	Stored as a String instead of datetime	Cannot perform time-series analysis	Convert to datetime
TotalSteps	Outliers	Contains extreme high/low values	Distorts statistical analysis	Remove using IQR
Sleep Disorder	Missing Values / Ambiguity	Contains "NaN" values	Reduces data consistency	Fill missing values using "None"

BEFORE & AFTER EVIDENCE

01 Blood Pressure — Split into Systolic & Diastolic

```
print(df[['Blood Pressure']].dtypes)
Blood Pressure object
dtype: object
```

→

```
Systolic int64
Diastolic int64
```

02 All Attributes — Remove Duplicates

```
df.duplicated().sum()
→ np.int64(54)
```

→

```
df.duplicated().sum()
→ np.int64(0)
```

03 Review Text — Text Preprocessing & Sentiment Analysis

BEFORE — RAW REVIEWS

#	ProductID	Helpfulness_Score	Rating	Review_Title	Text
0	CFB001E4K	1	5	Good Quality Dog Food	I have bought several of the Vitality canned d...
1	CFB00813G	0	1	Not as Advertised	Product arrived labeled as Jumbo Salted Peanut...
2	CFB000LQO	1	4	"Delight" says it all	This is a confection that has been around a fe...
3	CFB000UA0	3	2	Cough Medicine	If you are looking for the secret ingredient i...
4	CFB006K2Z	0	5	Great taffy	Great taffy at a great price. There was a wid...

↓

AFTER — CLEAN TEXT + SENTIMENT

#	ProductID	H_Score	Rating	Clean_Text	Sentiment_Label	Score
0	CFB001E4K	1	5	i have bought several of the vitality canned d...	**Positive**	**1**
1	CFB00813G	0	1	product arrived labeled as jumbo salted peanut...	**Negative**	**-1**
2	CFB000LQO	1	4	this is a confection that has been around a fe...	**Positive**	**1**
3	CFB000UA0	3	2	if you are looking for the secret ingredient i...	**Negative**	**-1**
4	CFB006K2Z	0	5	great taffy at a great price there was a wide...	**Positive**	**1**

04 Activity Date — Convert to datetime

```
print(df[['ActivityDate']].dtypes)
ActivityDate object
dtype: object
```

→

```
ActivityDate
datetime64[ns]
```

05 TotalSteps — Remove Outliers using IQR



06 Sleep Disorder — Fill NaN with "None"

```
df['Sleep Disorder']
.value_counts(dropna=False)

NaN 219
Sleep Apnea 78
Insomnia 77
```

→

```
None 219
Sleep Apnea 78
Insomnia 77
```

Lab 4: Schema Matching

COMPREHENSIVE SCHEMA MATCHING TABLE

SOURCE A ATTRIBUTE (DAILYACTIVITY_MERGED)	SOURCE B ATTRIBUTE (SLEEP_HEALTH_AND_LIFESTYLE)	MATCH TYPE	CONFIDENCE
Id	**Person ID**	Match	High
› Both serve as a unique identifier for individuals in datasets.			
TotalSteps	**Daily Steps**	Match	High
› Identical semantic meaning; both record the total steps taken daily.			
VeryActiveMinutes	**Physical Activity Level**	Partial Match	Medium
› Core of the physical activity level representing the high-intensity component.			
FairlyActiveMinutes	**Physical Activity Level**	Partial Match	Medium
› This attribute is a key constituent of the physical activity level, reflecting the moderate-intensity portion.			
LightlyActiveMinutes	**Physical Activity Level**	Partial Match	Medium
› These minutes serve as a supplementary part of the physical activity level, contributing to the total movement.			
SedentaryMinutes	**Physical Activity Level**	Partial Match	Medium
› Sedentary minutes represent inactivity, which is part of the overall physical activity profile.			

The table demonstrates the schema matching process by identifying equivalent attributes and relationships between heterogeneous datasets before integration into the data warehouse.

Lab 5: Schema Mapping

ATTRIBUTE MAPPING TABLE

SOURCE A ATTRIBUTE (DAILYACTIVITY_MERGED)	SOURCE B ATTRIBUTE (SLEEP_HEALTH_AND_LIFESTYLE)	MAPPING TYPE	SHORT DESCRIPTION
Id	**Person ID**	1:1	Direct mapping of the unique identifier from FitBit to the Sleep dataset.
TotalSteps	**Daily Steps**	1:1	Transferring the total steps recorded by the tracker to the health record.
VeryActiveMinutes, FairlyActiveMinutes, LightlyActiveMinutes	**Physical Activity Level**	N:1	Consolidating specific activity durations from FitBit into a single level in the Sleep dataset.

TRANSFORMATION RULES (SQL-STYLE)

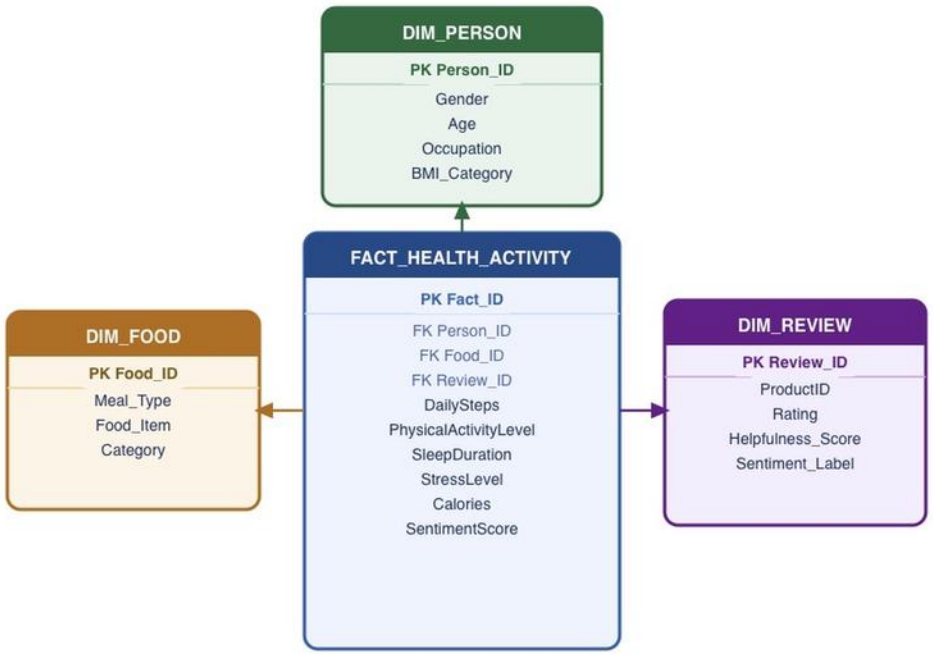
RULE #	RULE EXPRESSION (SQL-STYLE)	EXPLANATION
1	`SELECT Id AS Person_ID FROM dailyActivity_merged`	Maps the unique identifier **Id** to the target attribute **Person_ID**.
2	`SELECT TotalSteps AS Daily_Steps FROM dailyActivity_merged`	Maps the activity column **TotalSteps** to the health column **Daily_Steps**.
3	`SELECT CASE WHEN VeryActiveMinutes > 30 THEN 'High' ELSE 'Low' END AS Physical_Activity_Level`	Converts numeric activity minutes into a categorical **Activity Level (High/Low)**.

The tables demonstrate the schema mapping process by defining transformation rules and mapping equivalent attributes between heterogeneous datasets to unify the data structure for the data warehouse.

Lab 6: Star Schema & Data Warehouse & ETL

The star schema was designed to integrate lifestyle, nutrition, sleep, and activity datasets into a unified warehouse structure to support analytical queries and reporting.

STAR SCHEMA DIAGRAM



ETL PROCESS

E Extract

- Extracted health, nutrition, activity, and review datasets from multiple heterogeneous sources including Fitbit, Sleep Health, Food Nutrition, and Food Reviews datasets.

T Transform

- Cleaned missing values, duplicates, and outliers.
- Converted inconsistent data types into standardized formats.
- Applied schema matching and mapping.
- Transformed unstructured review text into structured sentiment features.

L Load

- Loaded the transformed and integrated data into the star schema warehouse structure to support SQL analysis, OLAP operations, and reporting.

Lab 7: Strategic Questions + OLAP / Data Mining

SQL QUERIES — BUSINESS QUESTIONS & OUTPUTS

Q1 Which meal type has the highest average calories?

SQL QUERY

```
SELECT DIM_FOOD.Meal_Type,
  AVG(FACT_HEALTH_ACTIVITY.Calories)
    AS Avg_Calories
FROM FACT_HEALTH_ACTIVITY
JOIN DIM_FOOD
  ON FACT_HEALTH_ACTIVITY.Food_ID
   = DIM_FOOD.Food_ID
GROUP BY DIM_FOOD.Meal_Type
ORDER BY Avg_Calories DESC;
```

OUTPUT

Identified the meal type with the highest average calorie intake

SQL

Meal_Type	Avg_Calories
Lunch	914.0000
Snack	852.9167
Breakfast	791.2941
Dinner	789.4167

Q2 How do stress levels and sleep duration vary across BMI categories?

SQL QUERY

```
SELECT DIM_PERSON.BMI_Category,
  MAX(StressLevel) AS Highest_Stress,
  MIN(StressLevel) AS Lowest_Stress,
  MAX(SleepDuration) AS Max_Sleep,
  MIN(SleepDuration) AS Min_Sleep
FROM FACT_HEALTH_ACTIVITY
JOIN DIM_PERSON
  ON FACT_HEALTH_ACTIVITY.Person_ID
   = DIM_PERSON.Person_ID
GROUP BY DIM_PERSON.BMI_Category;
```

OUTPUT

Compared stress levels and sleep duration among different BMI categories to identify health and lifestyle patterns.

SQL

BMI_Category	H_Stress	L_Stress	Max_Sleep	Min_Sleep
Overweight	6	6	6.10	6.10
Normal	58	0	10.20	0.60
Obese	8	7	6.30	5.90

Q3 Which meal type receives the most positive reviews?

SQL QUERY

```
SELECT DIM_FOOD.Meal_Type,
  COUNT(DIM_REVIEW.Sentiment_Label)
    AS Positive_Reviews
FROM FACT_HEALTH_ACTIVITY
JOIN DIM_FOOD
  ON FACT_HEALTH_ACTIVITY.Food_ID
   = DIM_FOOD.Food_ID
JOIN DIM_REVIEW
  ON FACT_HEALTH_ACTIVITY.REViewID
   = DIM_REVIEW.Review_ID
WHERE DIM_REVIEW.Sentiment_Label = 'Positive'
GROUP BY DIM_FOOD.Meal_Type
ORDER BY Positive_Reviews DESC;
```

OUTPUT

Identified the meal type with the highest number of positive reviews

SQL

Meal_Type	Positive_Reviews
Breakfast	13
Snack	11
Dinner	11
Lunch	9

Q4 Which occupation has the highest total calories and daily steps?

SQL QUERY

```
SELECT DIM_PERSON.Occupation,
  SUM(FACT_HEALTH_ACTIVITY.Calories)
    AS Total_Calories,
  SUM(FACT_HEALTH_ACTIVITY.DailySteps)
    AS Total_Steps
FROM FACT_HEALTH_ACTIVITY
JOIN DIM_PERSON
  ON FACT_HEALTH_ACTIVITY.Person_ID
   = DIM_PERSON.Person_ID
GROUP BY DIM_PERSON.Occupation
ORDER BY Total_Steps DESC;
```

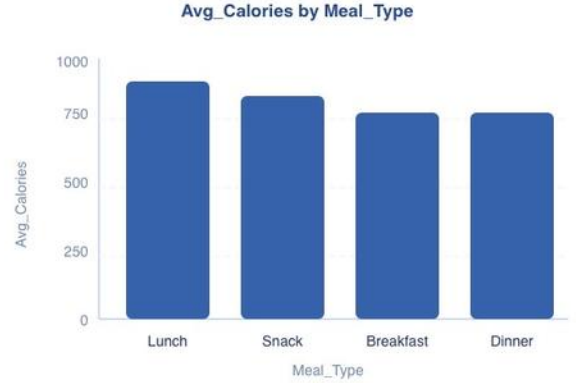
OUTPUT

Identified occupations with the highest total calorie consumption and physical activity levels.

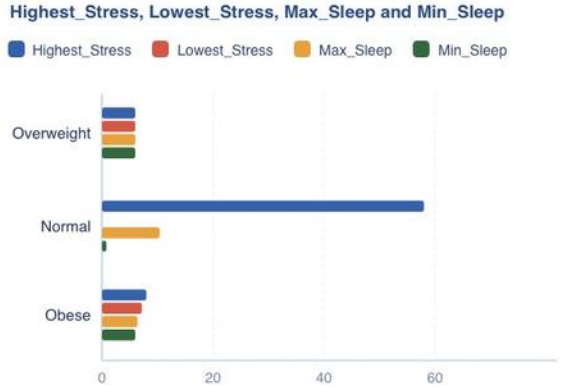
SQL

Occupation	Total_Calories	Total_Steps
Doctor	34593	306507
Engineer	4228	20805
Nurse	1605	11279
Software Engineer	275	7200
Sales Representative	455	6000
Teacher	230	3500

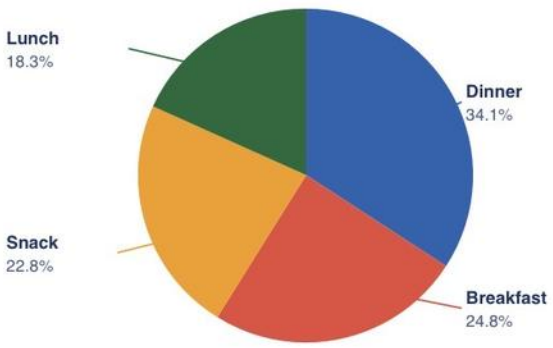
Key Insights + Visual Dashboards & Interactive Reports



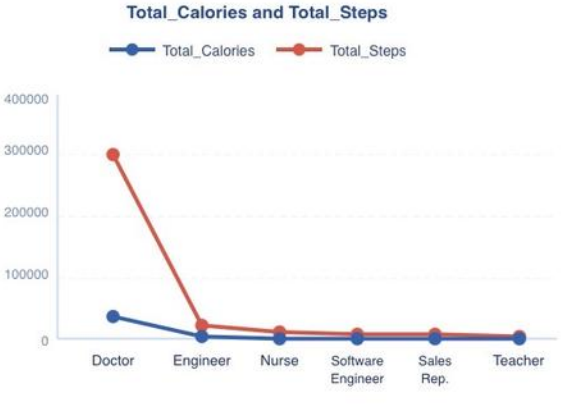
Presents a visual representation of the average calories for each meal type



Presents a visual representation of stress levels and sleep duration across different BMI categories



Presents a visual representation of the number of positive reviews for each meal type



Presents a visual representation of total calories and daily steps across different occupations

Final Recommendations & Lessons Learned

Final Recommendations

- Encourage healthier nutrition and sleep habits.
- Use integrated analytics to monitor lifestyle behavior.
- Improve physical activity awareness through health analysis.
- Expand the warehouse using additional healthcare datasets.
- Apply machine learning models for future health prediction.

Lessons Learned

- Data cleaning significantly improves data quality and analysis accuracy.
- Schema matching and mapping are essential for integrating heterogeneous datasets.
- ETL processes help organize and standardize complex data sources.
- SQL and OLAP techniques support strategic analytical decision-making.
- Sentiment analysis transforms unstructured reviews into meaningful structured data.
- Visual dashboards improve understanding of health and lifestyle patterns.